<u>Celebal Summer Internship 2026</u>

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

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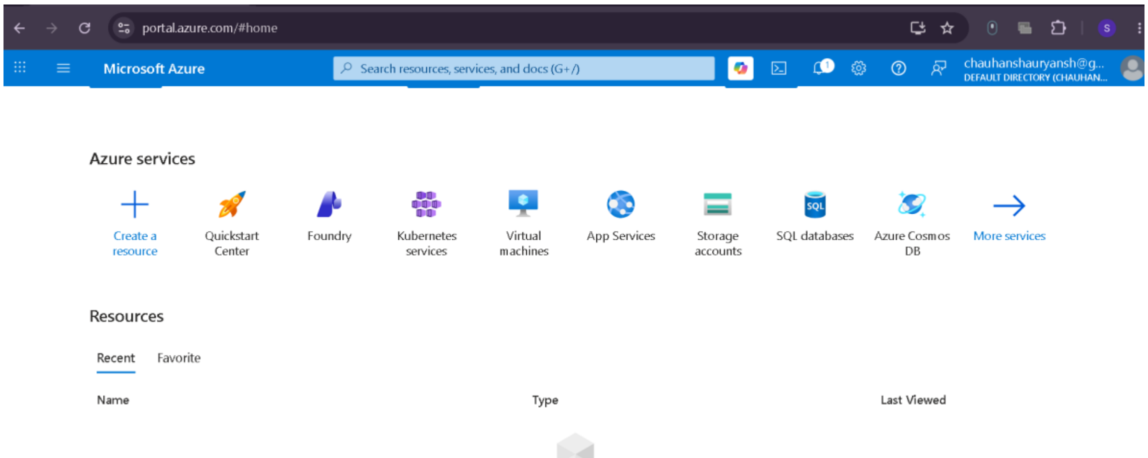
Course: B.Tech Computer Science & Engineering

Assignment of week 4

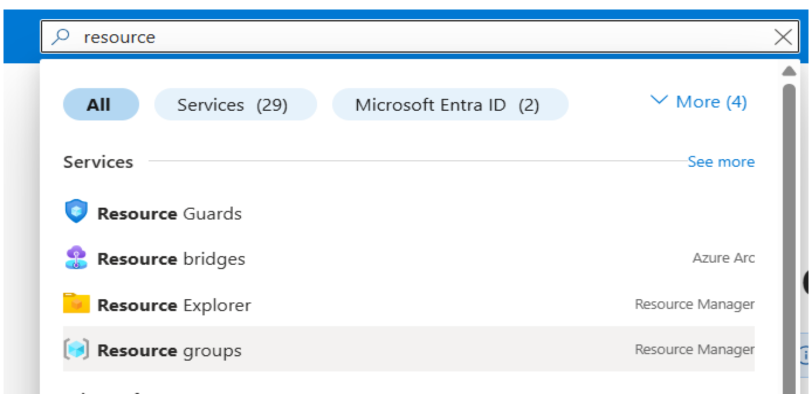
Phase no. 1: Foundation & Storage Setup

Step 1: Create a Resource Group A Resource Group acts as a logical container for all your project assets.

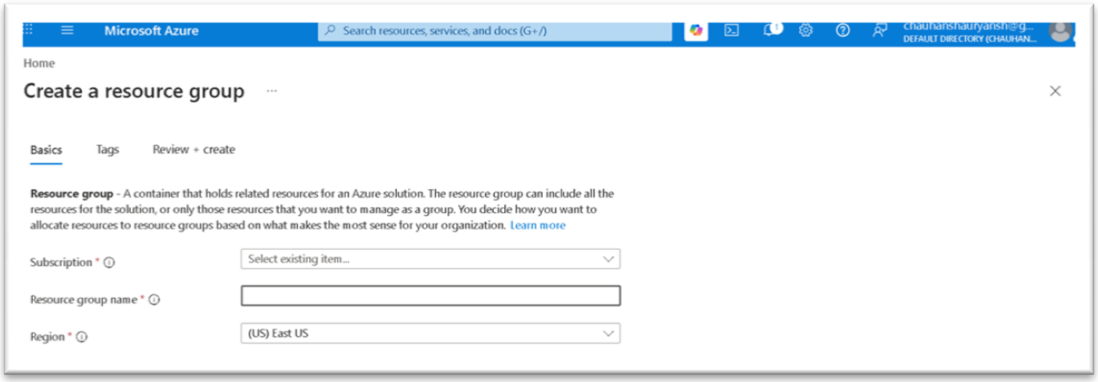
1. Log in to the Azure Portal.



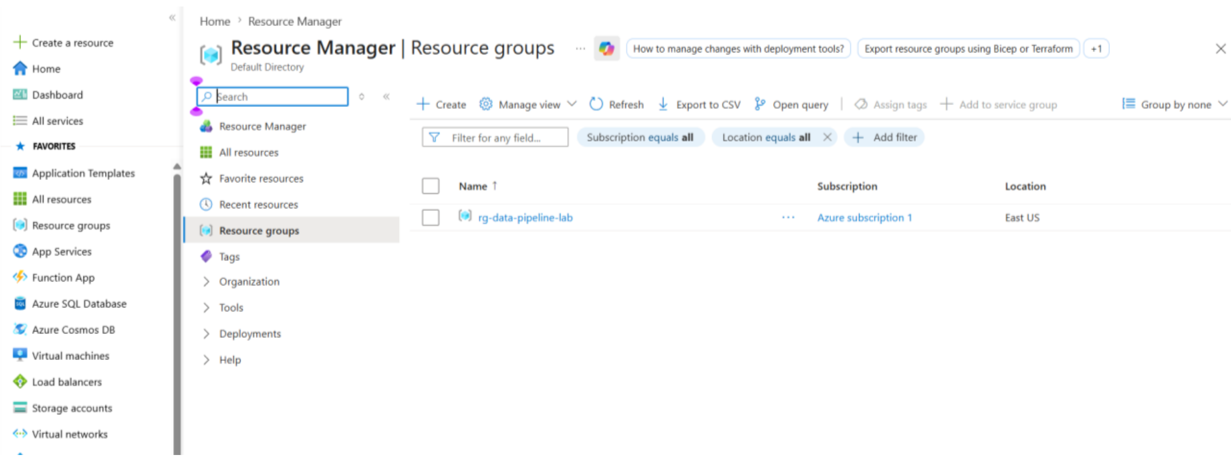
2. In the search bar, type **Resource groups** and select it.



3. Click **+ Create**.Select your Subscription, name your Resource Group (e.g., rg-data-pipeline-lab), and choose a region (e.g., East US).

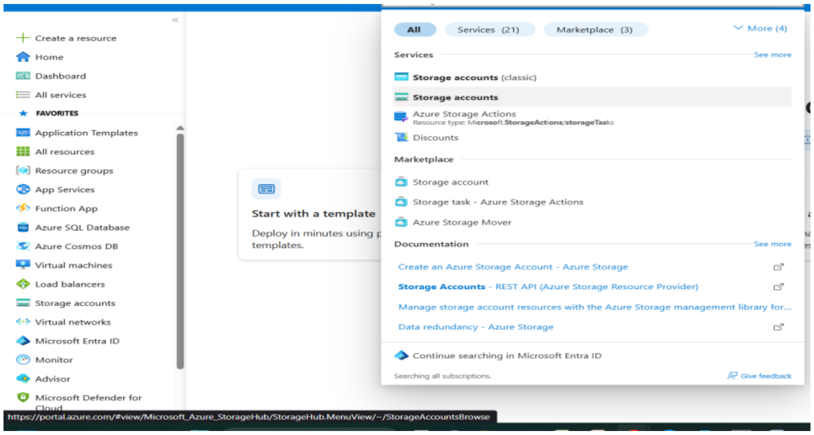


4. Click **Review + create**, then **Create**.

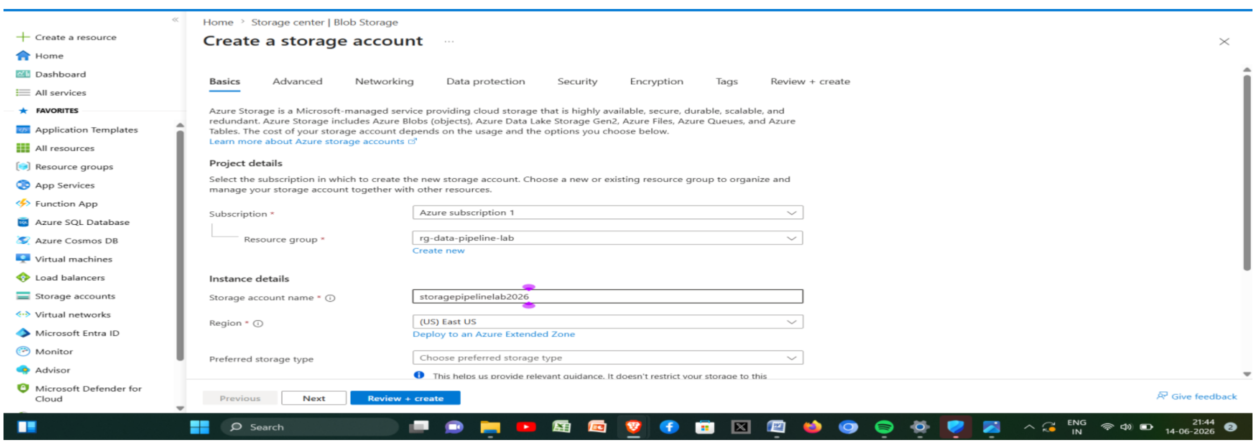


Step 2: Create a Storage Account & Blob Container This will act as both the source and destination for your data.

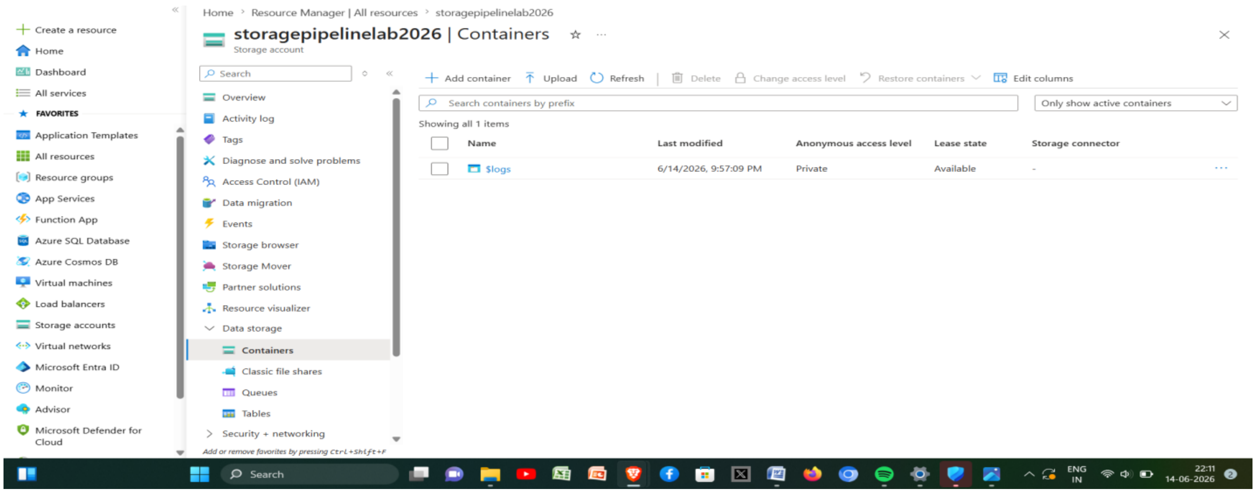
1. Search for **Storage accounts** and click **+ Create**.



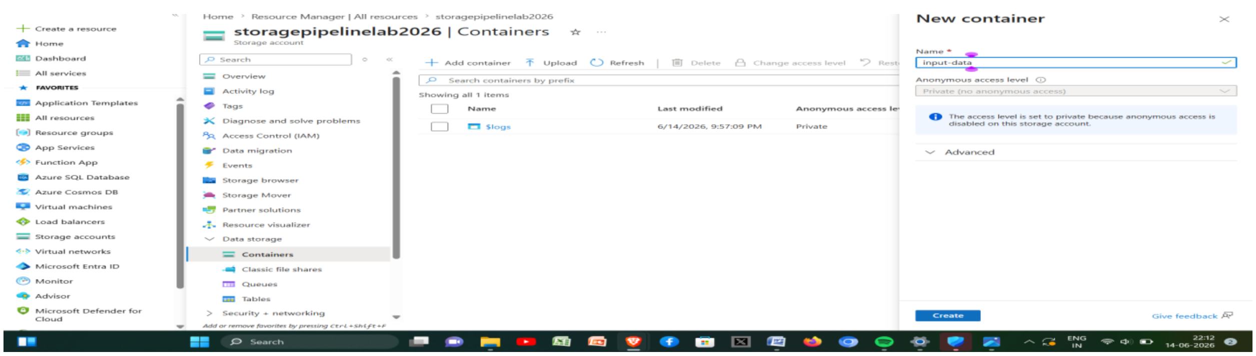
2. Select your new Resource Group and give the storage account a globally unique name (lowercase letters and numbers only, e.g., storagepipelinelab2026). Leave other settings as default and create it.

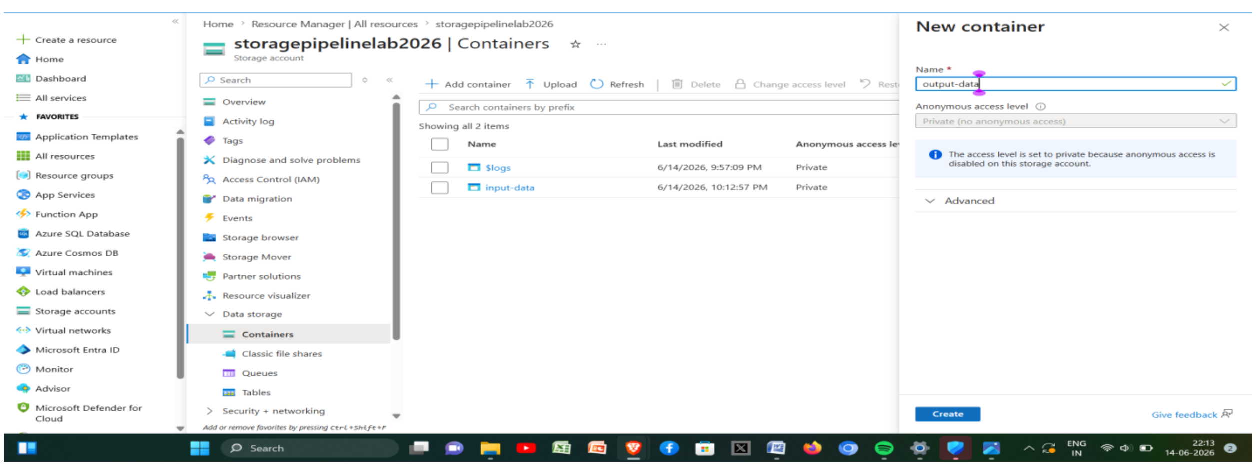


3. Once deployed, go to the resource. On the left menu under **Data storage**, click **Containers**.

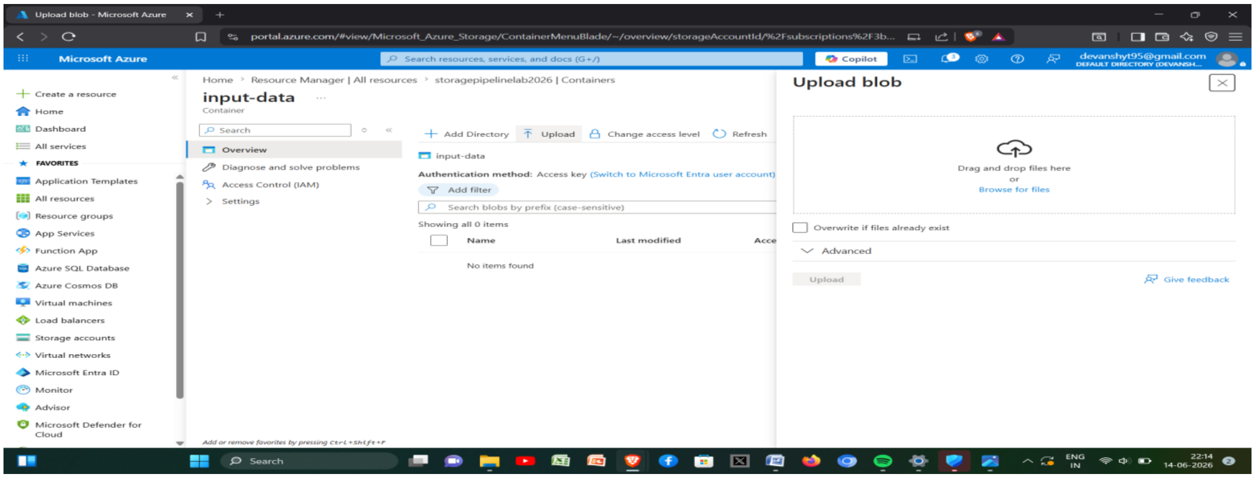


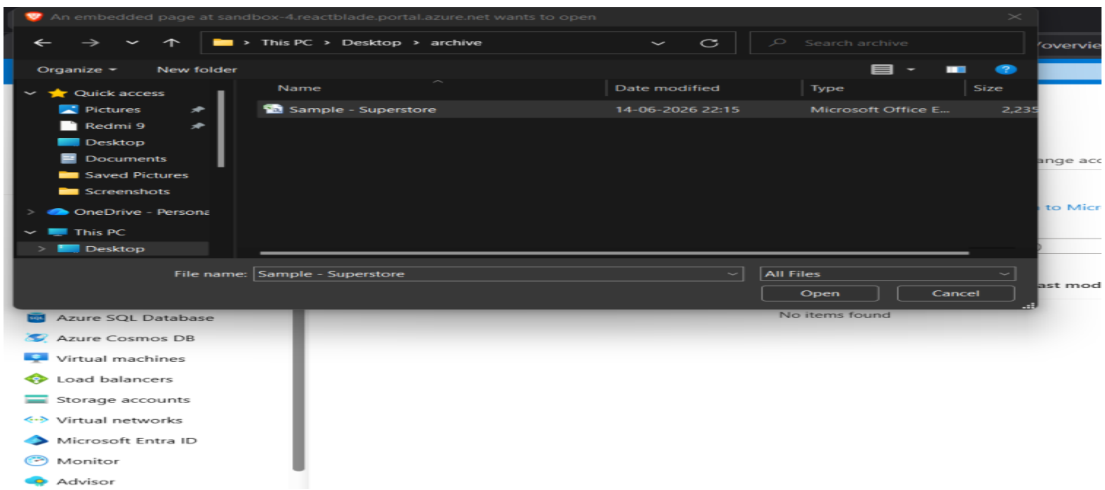
4. Create two containers: click **+ Container**, name one input-data, and repeat to create output-data.

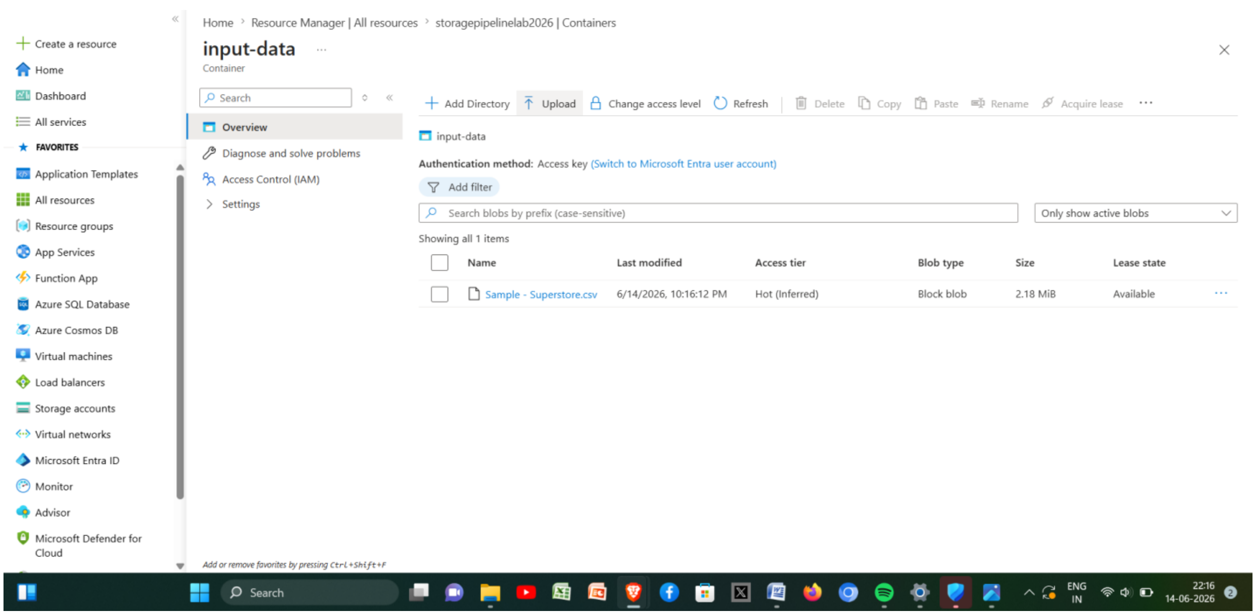




5. Click into input-data, click **Upload**, and select a simple CSV file from your local machine.



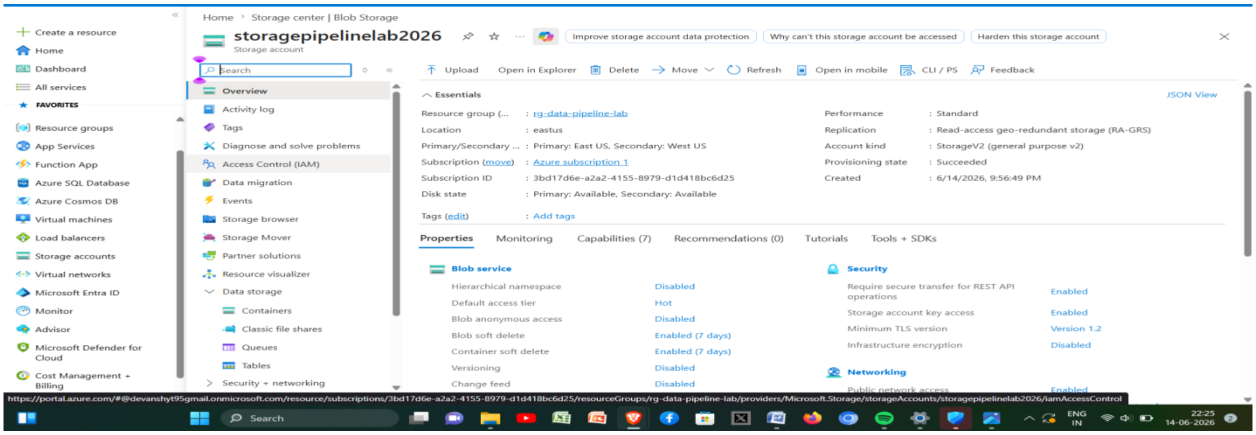




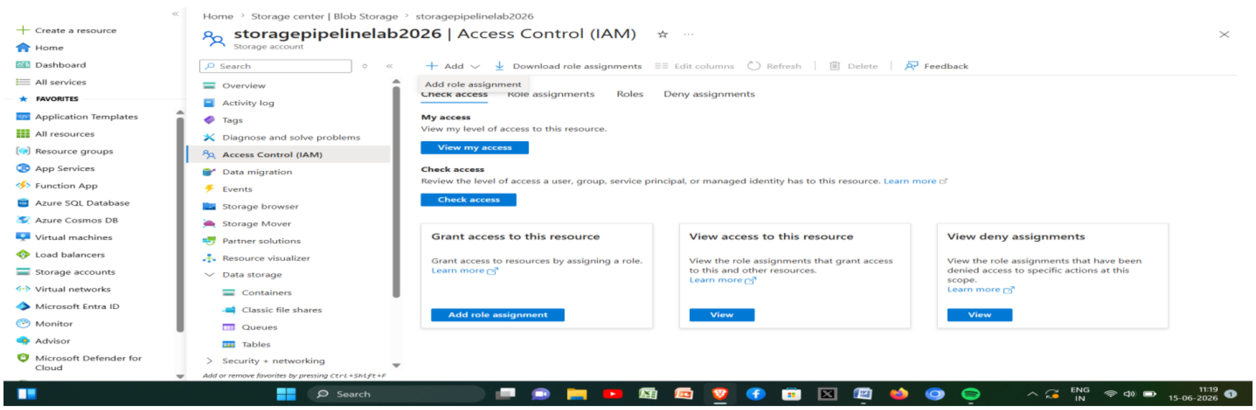
Phase no. 2: Security & IAM Roles

Step 3: Assign IAM Roles To allow Azure Data Factory to read and write to your storage account securely, you need to grant it permissions using Role-Based Access Control (RBAC).

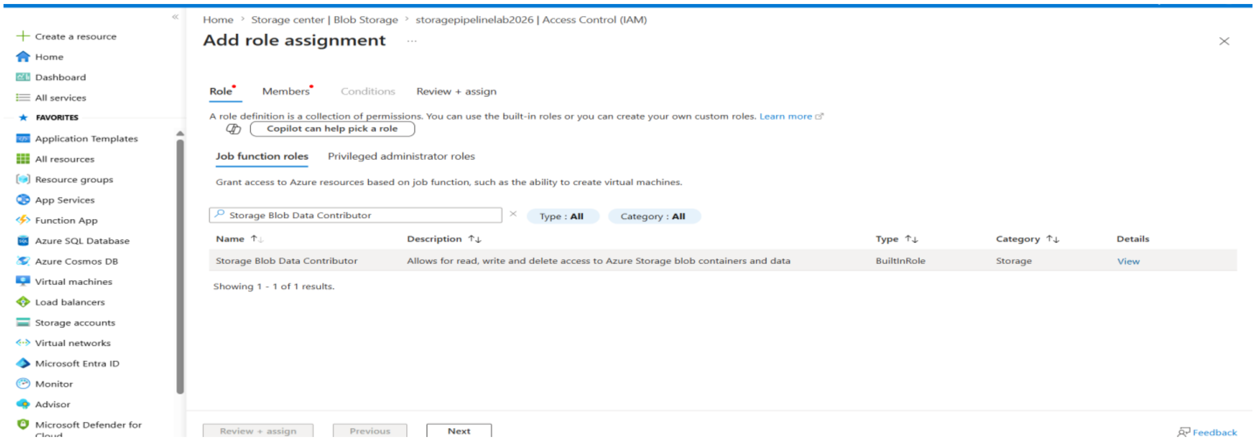
1. Navigate back to your **Storage Account** overview.
2. Click on **Access Control (IAM)** in the left menu.



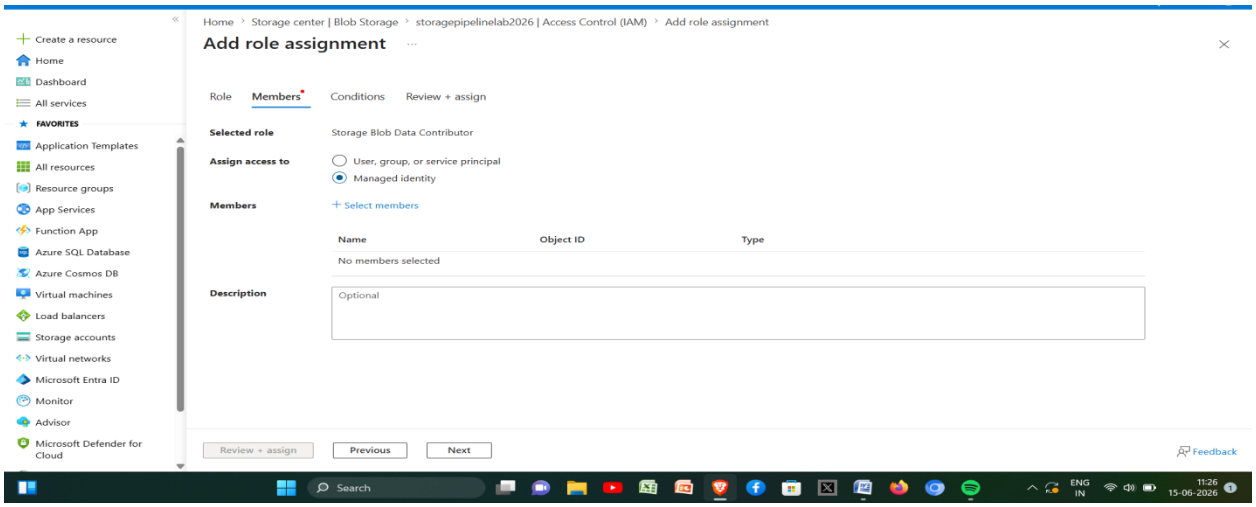
3. Click **+ Add** -> **Add role assignment**.



4. Search for the role **Storage Blob Data Contributor** (this is better than just "Contributor" as it allows reading/writing the actual data inside the blobs).



5. Assign access to **Managed identity**.

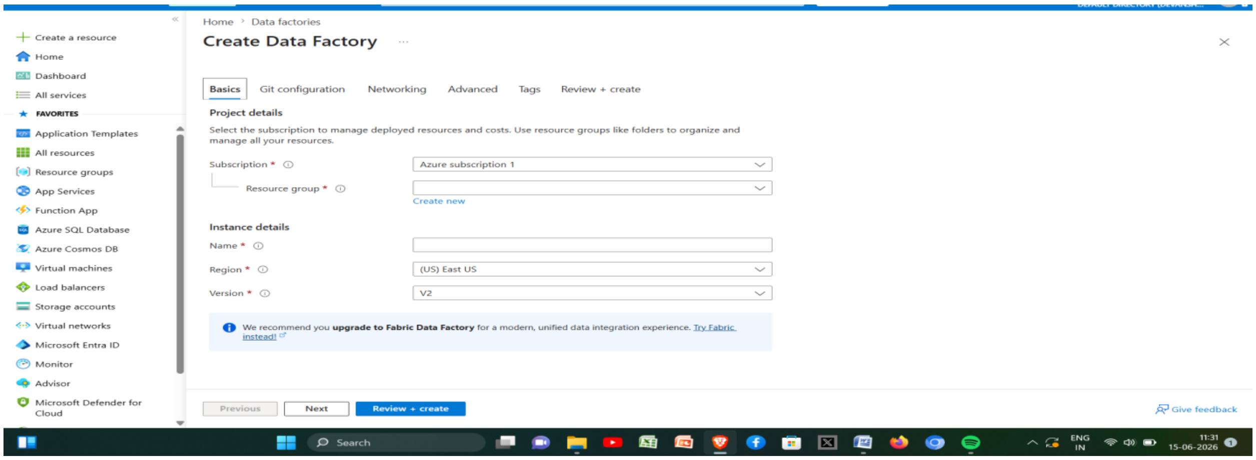


6. *Note: You will finish assigning this to your specific Data Factory in Step 4 after you create it.*

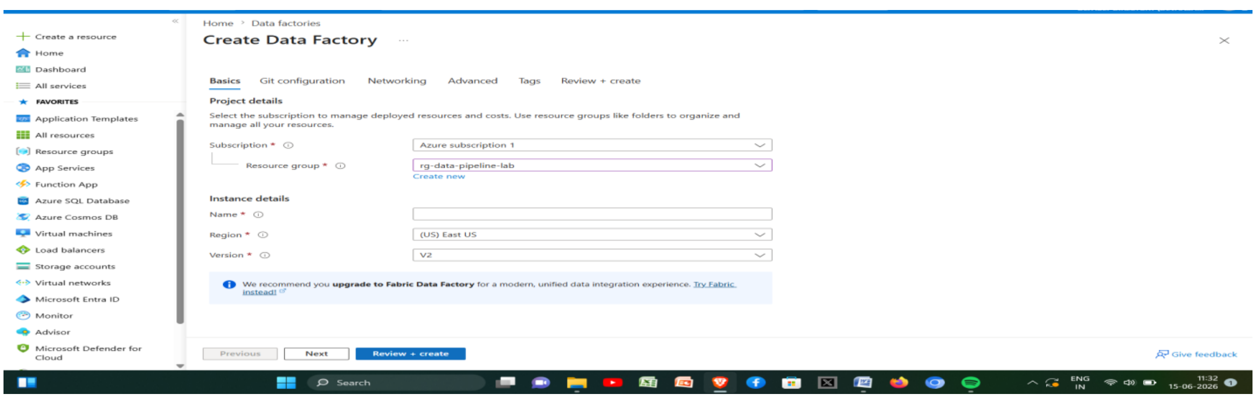
Phase no.3: Azure Data Factory (ADF) Setup

Step 4: Provision ADF and Explore the UI

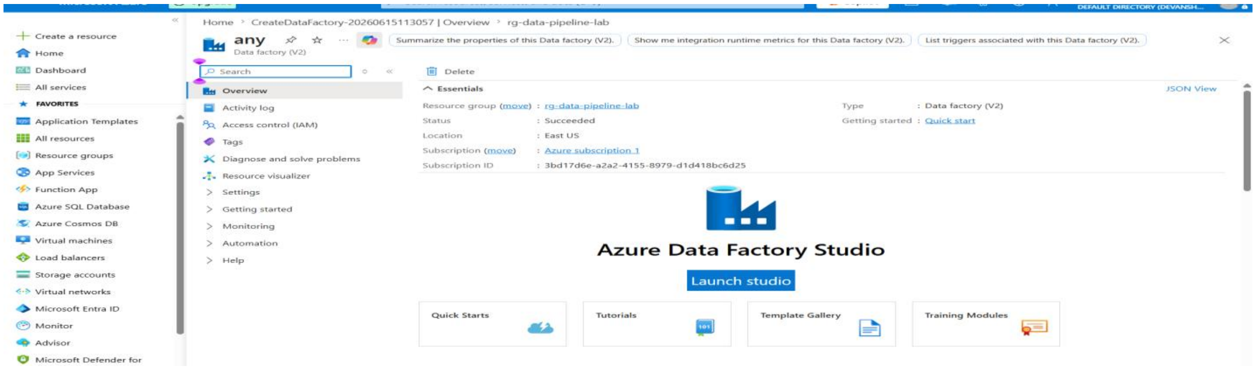
1. Search for **Data factories** in the portal and click **+ Create**.



2. Select your Resource Group, name your ADF (e.g., adf-pipeline-lab), and click **Review + create**, then **Create**.

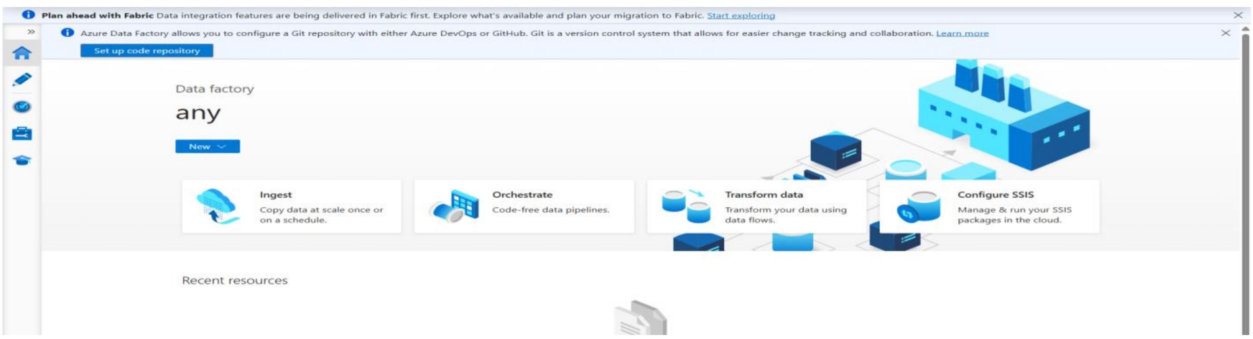


3. Once deployed, go to the resource and click **Launch studio**

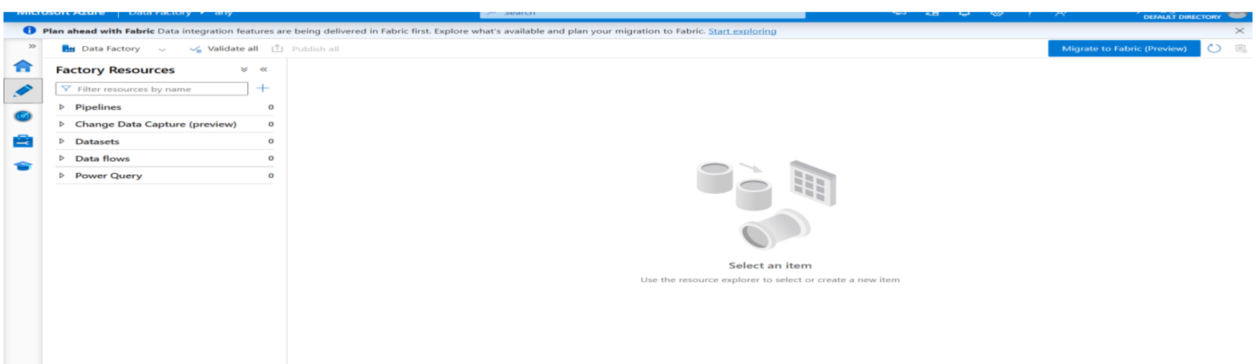


4. **Explore the UI:**

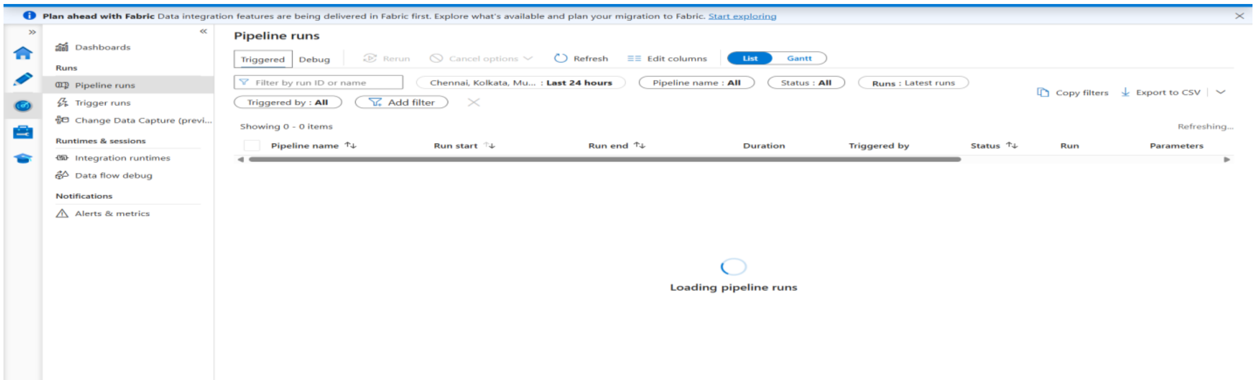
Home: Quick starts and templates.



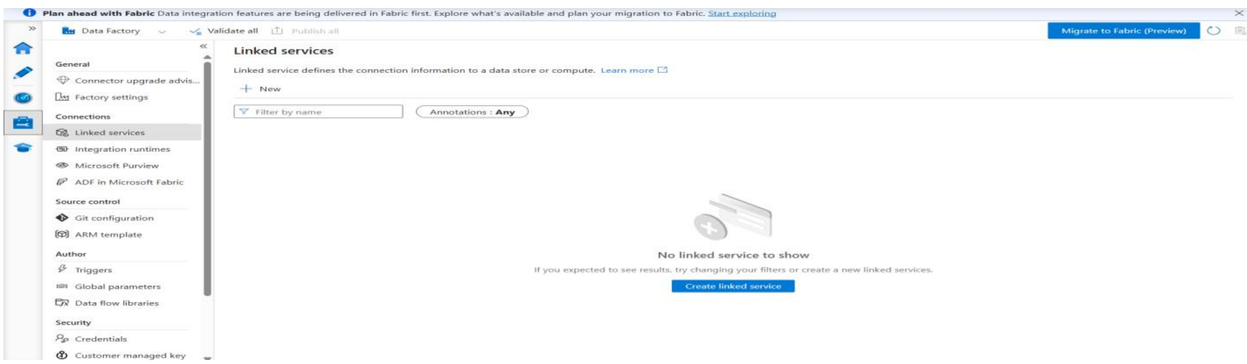
Author (Pencil icon): Where you build pipelines, datasets, and data flows.



Monitor (Gauge icon): Where you check the execution status of your pipeline runs.



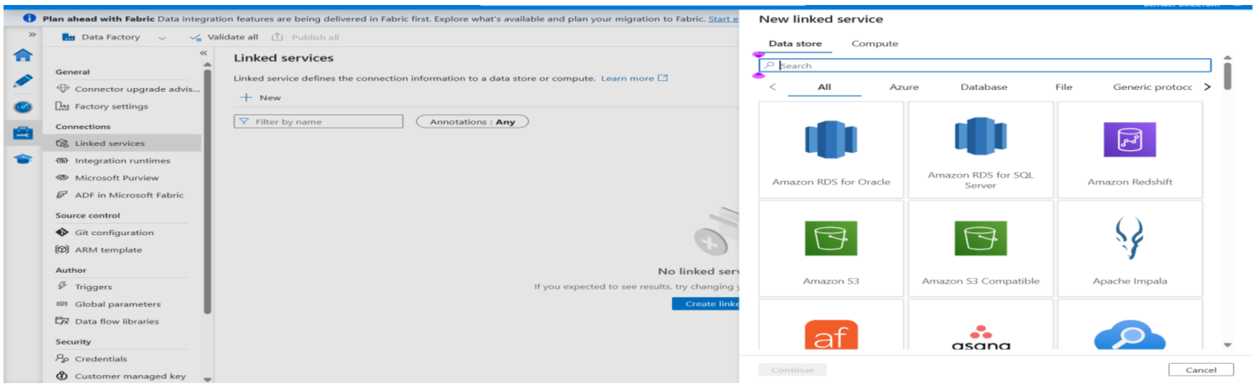
Manage (Toolbox icon): Where you configure Linked Services (connections) and integration runtimes.



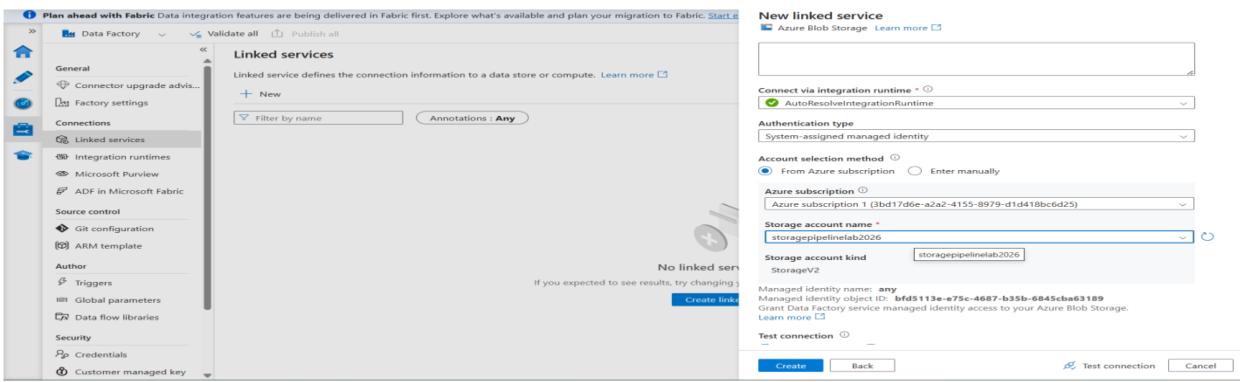
Phase no. 4: Linked Services & Datasets

Step 5: Create the Linked Service This is the connection string between ADF and your Storage Account.

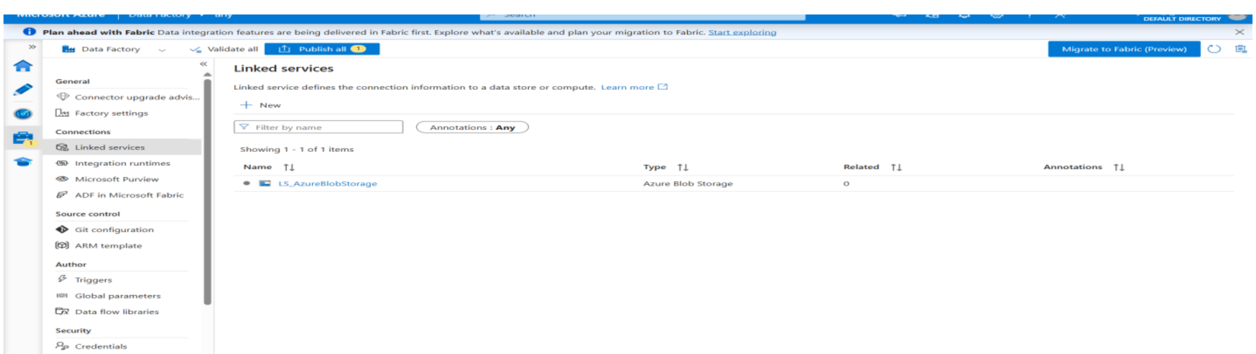
1. In ADF Studio, go to the **Manage** tab -> **Linked services** -> **+ New**.



2. Select **Azure Blob Storage** and click Continue.
3. Under Authentication method, select **System Assigned Managed Identity** (which utilizes the IAM role you set up).

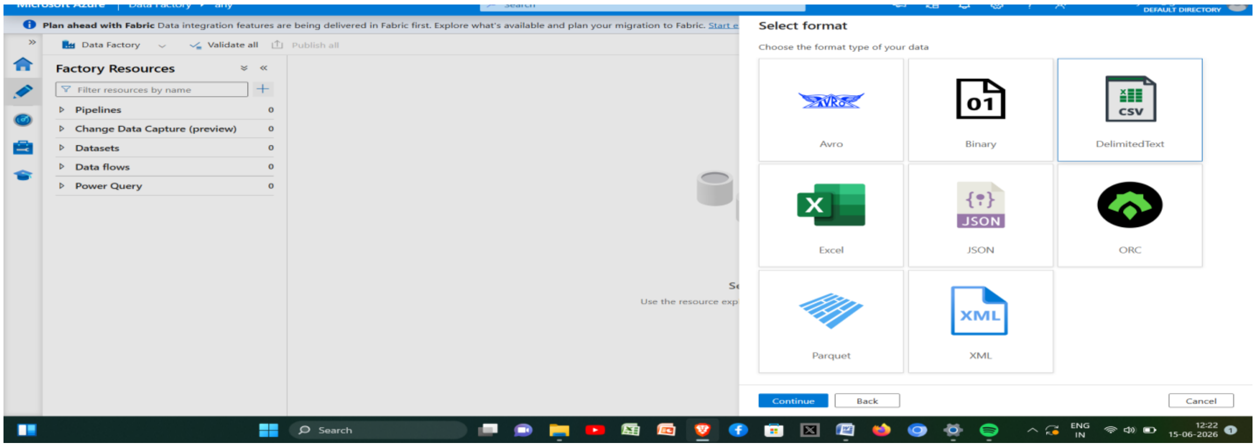


4. Select your Azure subscription and your specific Storage account name. Click **Test connection**, then **Create**.

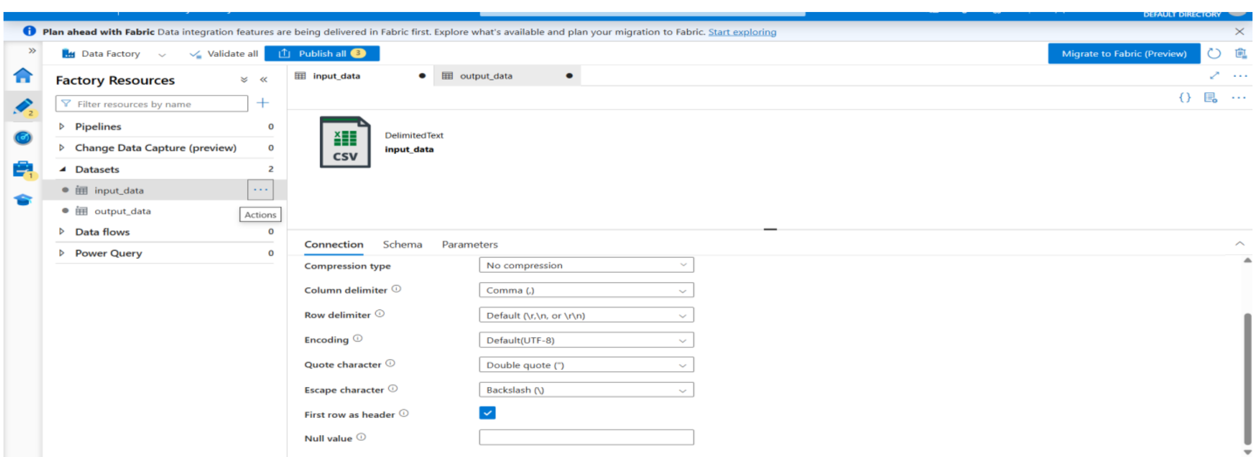


Step 6: Create Source and Destination Datasets Datasets represent the specific data structures you want to interact with.

1. Go to the **Author** tab -> Click the + icon -> **Dataset**.
2. Select **Azure Blob Storage** -> **DelimitedText** (CSV).



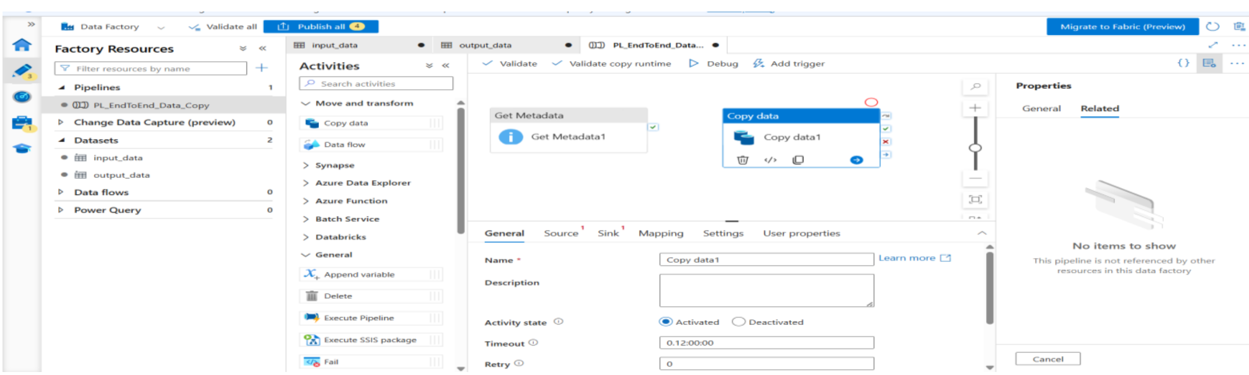
3. **Source Dataset:** Name it DS_Input_CSV. Select your LS_AzureBlobStorage linked service. Browse to the file path: input-data/yourfile.csv. Check "First row as header". Click OK.
4. **Destination Dataset:** Repeat the process. Name it DS_Output_CSV. Select the same linked service. For the file path, only select the output-data container (leave the filename blank so it auto-generates or retains the source name). Click OK.



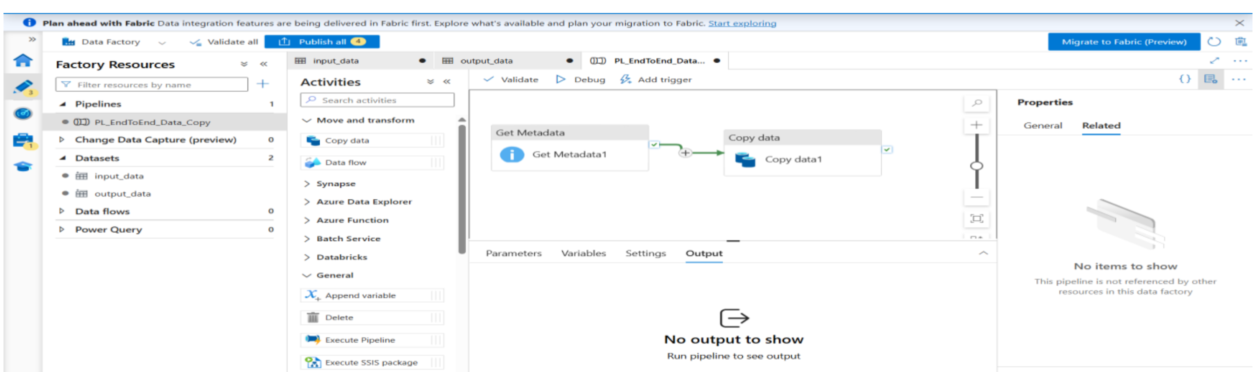
Phase no.5: Building the Pipeline

Step 7: Get Metadata and Copy Data Activities

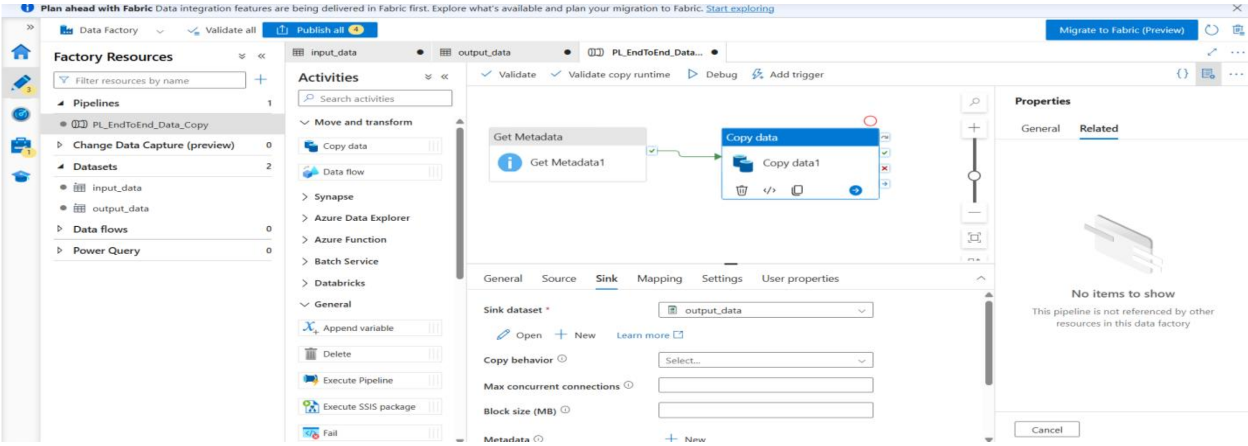
1. In the **Author** tab, click + -> **Pipeline** -> **Pipeline**. Name it PL_EndToEnd_Data_Copy.
2. Expand the **General** section in the Activities pane, drag **Get Metadata** onto the canvas.
   - Go to the **Dataset** tab of this activity and select DS_Input_CSV.
   - Go to the **Field list** tab, click **+ New**, and select **Item name** and **Exists**. (This validates the file is actually there before trying to copy it).



3. Expand the **Move & transform** section, drag **Copy data** onto the canvas.
4. Drag the green success box (the green checkmark) from the Get Metadata activity and connect it to the Copy data activity.



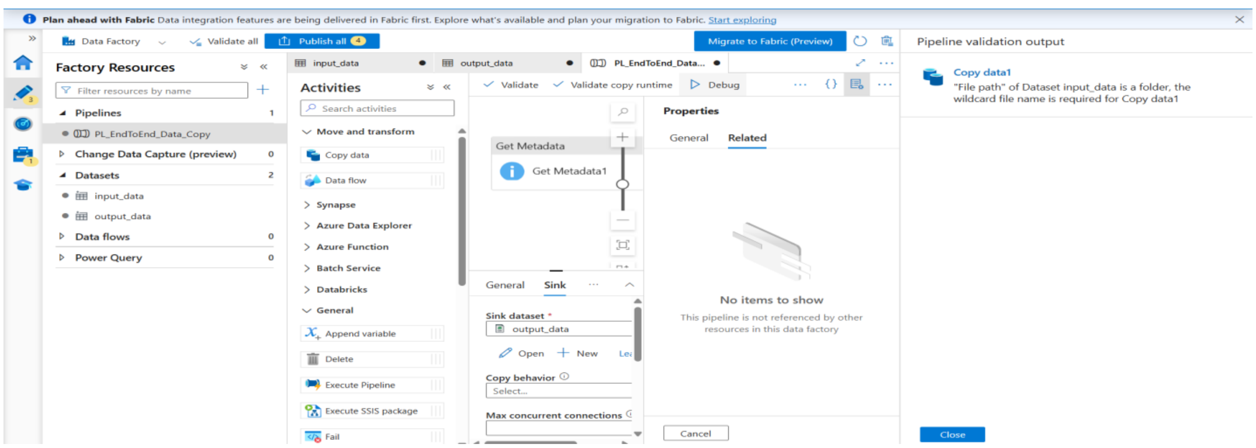
5. Configure **Copy data**:
   - **Source tab:** Select DS_Input_CSV.
   - **Sink tab:** Select DS_Output_CSV.



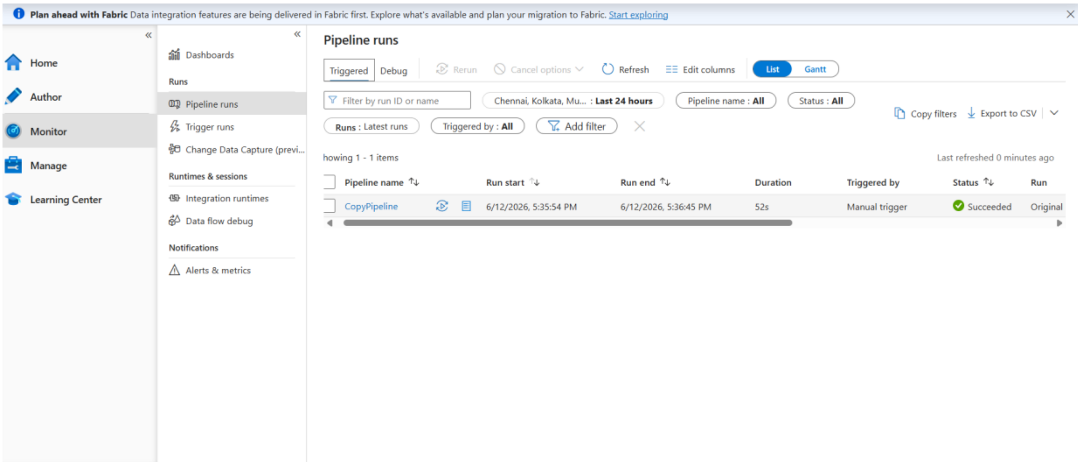
Phase 6: Execution & Monitoring

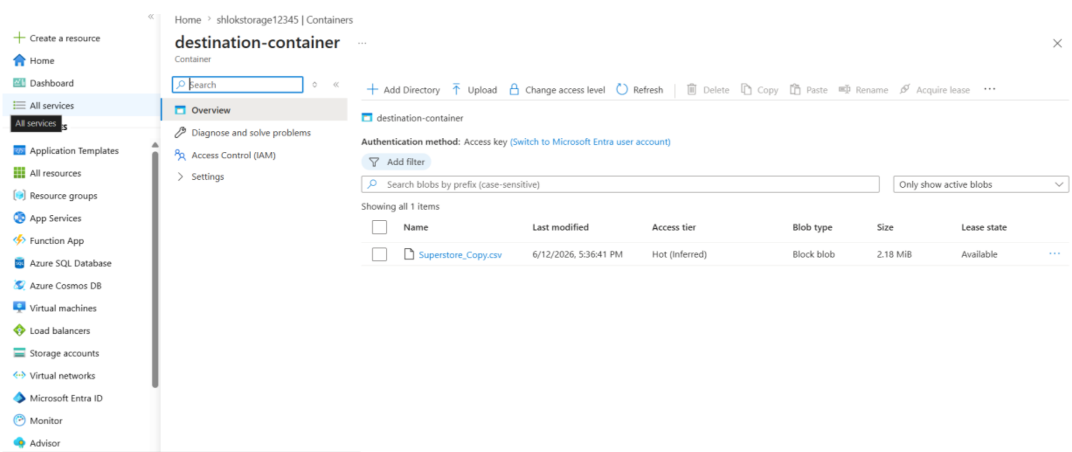
Step 8: Debug and Monitor

1. Click the **Debug** button at the top of the pipeline canvas. This executes the pipeline immediately without publishing.
2. Watch the Output tab at the bottom. You should see "Get Metadata" succeed, followed by "Copy Data" succeeding.



3. To verify, go to the **Monitor** tab on the far left. View the pipeline runs to see detailed execution stats.
4. Finally, go back to your Azure Portal -> Storage Account -> output-data container and verify the CSV file successfully arrived.





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Final output summary :

An end-to-end data pipeline was successfully orchestrated using Azure Data Factory. A Resource Group was established to house an Azure Storage Account and an ADF instance. Secure connectivity was achieved by assigning the 'Storage Blob Data Contributor' IAM role to the ADF's Managed Identity. Within ADF, a Linked Service was configured to establish the connection, and Delimited Text datasets were defined for both the source (input-data container) and destination (output-data container). The pipeline logic utilized a 'Get Metadata' activity to successfully validate the presence of the source CSV file, which sequentially triggered a 'Copy Data' activity upon success. The pipeline was executed via the Debug trigger, and the Monitor console verified a successful run, resulting in the CSV data being successfully written to the destination container.